Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

Title: Sales Performance Analysis Across Products, Categories, and Regions using Power BI Visualizations

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Sales Performance Analysis Across Products, Categories, and Regions using Power BI Visualizations

GitHub-Repo-Link: [prakruthidevanga/Business-Analytics: Business Analytics]()

Problem Statement

The management of a retail organization wants to quickly understand how sales and profit are performing across different products, categories, and regions to identify key business patterns and support data-driven decision-making. As a Data Analyst, the task is to import the given business dataset (Sample Superstore dataset) into Power BI, prepare it for analysis, and design appropriate visualizations that highlight sales performance, profitability, and regional trends.

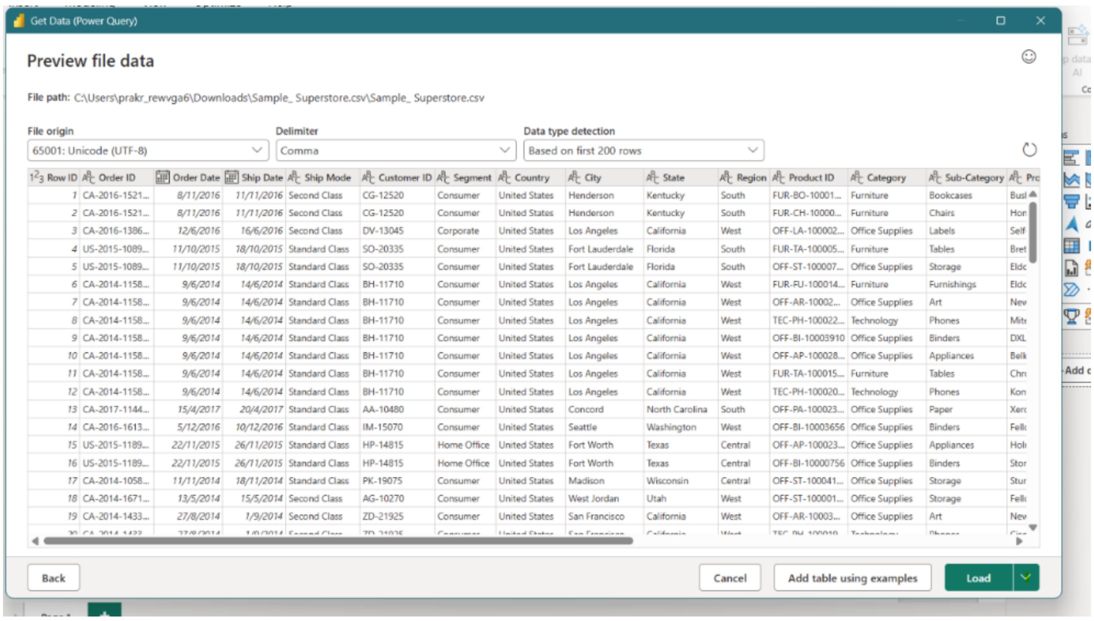
Methodology / Steps Followed

Step 1: Data Collection

- Obtained the Sample Superstore dataset (SampleSuperstore.csv) from Kaggle.
- The dataset contains 9,994 records with 13 fields, including Ship Mode, Segment, Region, Category, Sub-Category, Sales, Quantity, Discount, and Profit.
- Kaggle Link: [Sample Superstore]()

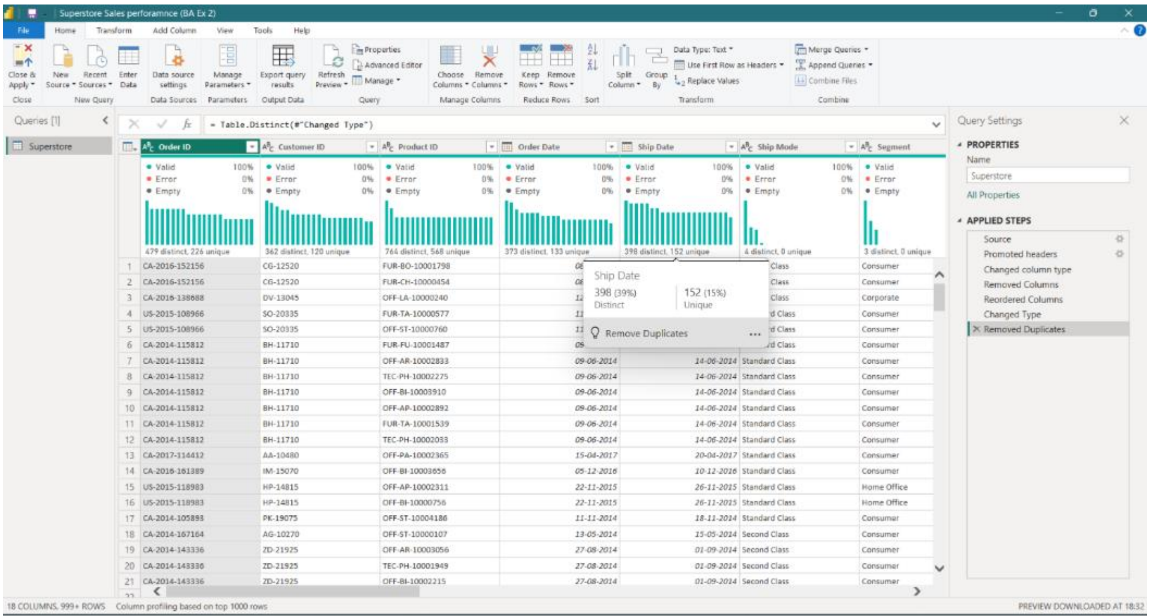
Step 2: Data Import

- Opened Power BI Desktop.
- Imported the dataset using Get Data → Text/CSV.
- Loaded the Sample Superstore CSV file into Power BI.



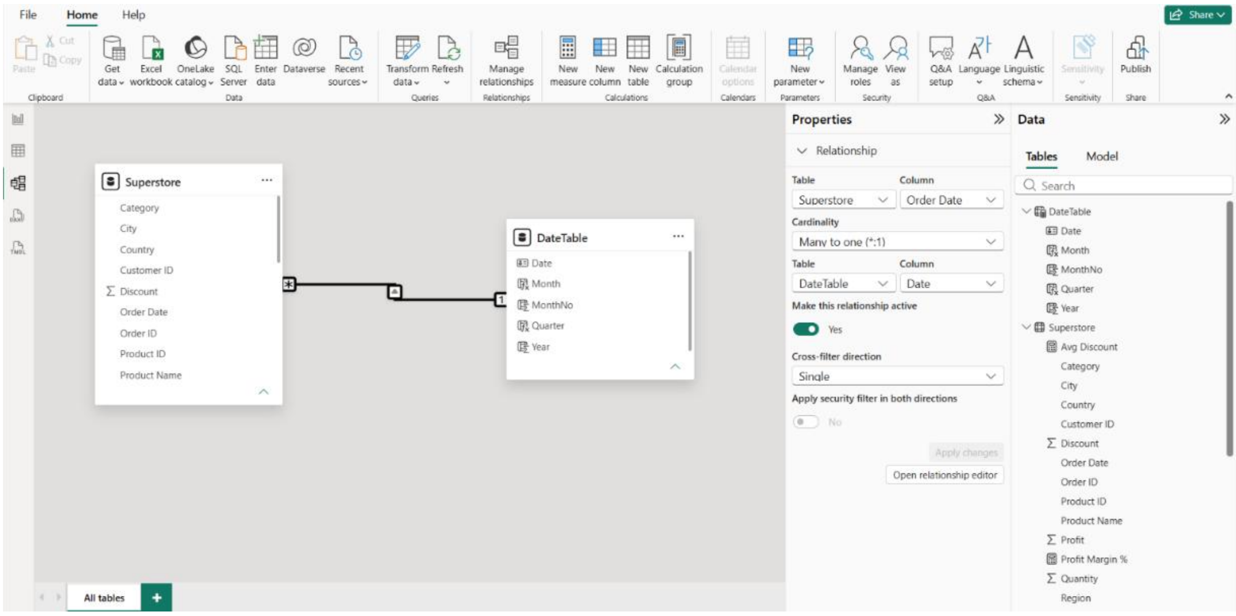
Step 3: Data Cleaning and Transformation

- Verified and corrected data types (Sales, Profit, and Discount set as Decimal Number; Quantity and Postal Code set as Whole Number).
- Selected all the columns and removed duplicate records using Power Query Editor.
- Confirmed there were no empty values in key fields (Sales, Profit, Category, Region).
- Renamed the query and columns appropriately for clarity.



Step 4: Data Modeling

- Created a Date table using DAX CALENDAR ().
- Added Year, Month, Month Name, Quarter columns.
- Marked it as an official Date Table.
- Built a relationship: Date table [Date] - one to many - Superstore [Order Date].
- Verified relationship in Model view (correct cardinality).



Step 5: Dax Measures

Created DAX measures

- Total Sales = SUM (Superstore [Sales])
- Total Profit = SUM (Superstore [Profit])
- Profit Margin % = DIVIDE ([Total Profit], [Total Sales], 0)
- Total Orders = DISTINCTCOUNT (Superstore [Order ID])
- Average Discount = AVERAGE (Superstore [Discount])
- Total Quantity = SUM (Superstore [Quantity])
- Sales Prior Year = CALCULATE ([Total Sales], SAMEPERIODLASTYEAR(DateTable[Date]))
- Sales YoY % = DIVIDE ([Total Sales] - [Sales Prior Year], [Sales Prior Year], 0)

These DAX measures were created to convert raw transactional data into consistent, reusable business calculations that power every KPI card, chart, and table across the report.

Step 6: Dashboard Development

- Created KPI cards for Total Sales, Total Profit, Profit Margin %, and Total Orders.
- Developed a bar chart for Sales by Category and Sales by Sub-Category.
- Developed a map visualization for Sales by Region/State.
- Created a scatter chart comparing Discount vs Profit to study the impact of discounting.
- Added a donut chart for Sales by Segment.
- Added slicers for Region, Category, and Segment to make the dashboard interactive.

Output

The Power BI dashboard was developed using the Sample Superstore dataset and provides the following key business insights:

Executive Overview

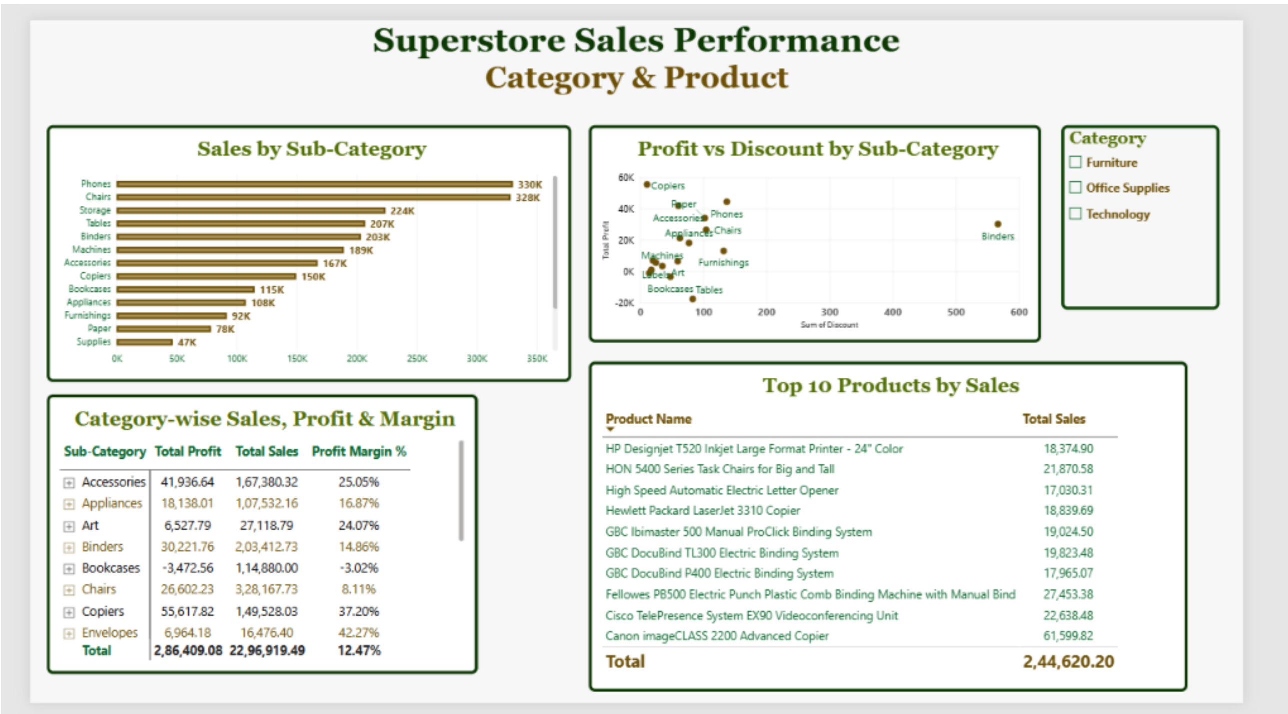
This page acts as the entry point of the report designed to answer, **"How is the business doing overall?"** before drilling into category-level or region-level detail on the following pages.

- Four KPI cards give a quick snapshot of performance.
- A line chart shows how sales and profit have moved month to month, so seasonal spikes are easy to spot.
- The map highlights which states are selling the most (and least).
- Slicers for Year, Category, Region, and Segment let anyone filter the whole page without needing a separate report.

Category & Product Performance

This page breaks down business performance at the **product and sub-category level**, going deeper than the company-wide view shown on the Overview page.

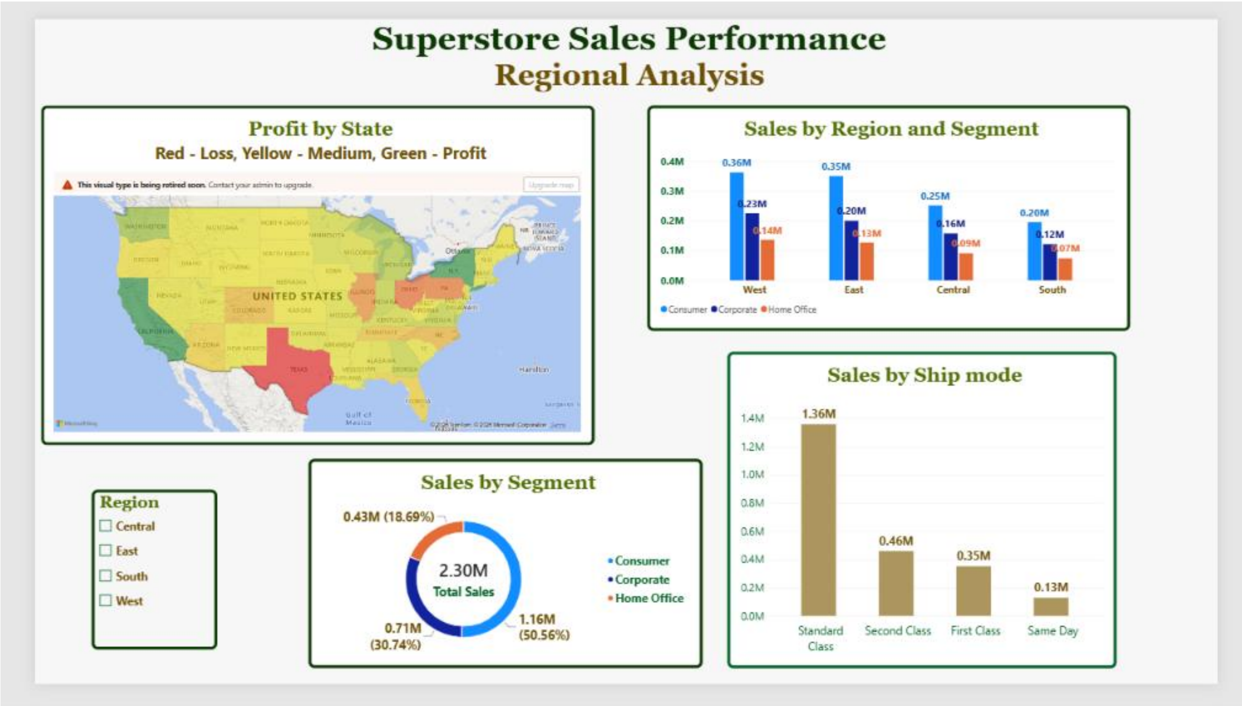
- The bar chart shows which sub-categories are pulling their weight and which aren't.
- The scatter chart is the interesting one — it shows where heavy discounting is losing profit.
- A table lays out Sales, Profit, and Margin % for every category and sub-category side by side.
- A Top 10 table shows the best-selling products, useful for stocking and marketing decisions.



Regional Analysis

This page focus from products to **geography and customer segments**, showing where the business performs well and where it doesn't.

- The map (Red for loss, Green for profit) makes it obvious which states are losing money.
- A column chart breaks down sales by region and customer segment.
- A bar chart shows which shipping method gets used most.
- A donut chart shows which customer segment brings in the most revenue.



Key Insights & Findings

- Sales clearly spike around November–December, likely holiday shopping.
- A handful of sub-categories account for most of the revenue.
- Tables and Bookcases sell well but barely make money (or lose money) because of steep discounts.
- The more something gets discounted, the less profit it makes — a consistent pattern.
- A few states show decent sales but still end up in the red.
- West and East regions are generally the strongest performers; Central lags behind.
- Consumer buyers bring in the most sales, but Corporate and Home Office tend to be more profitable per sale.
- Most orders ship Standard Class, which matters for delivery costs.
- Comparing sales year-over-year helps tell real growth apart from just seasonal ups and downs.
- Profit margin is basically a quick pulse-check on how healthy the business really is.

Conclusion

- The dataset was successfully cleaned, modeled, and turned into a working Power BI dashboard.
- Setting up a proper Date table made time-based analysis much more reliable.
- Using DAX measures kept every number on the report consistent instead of recalculating things differently each time.
- Splitting the report into three pages made it easy to look at the business from different angles overall, by product, and by region.
- The biggest takeaway is the discount-vs-profit relationship that's the kind of insight that changes business decisions.
- Overall, this shows how raw spreadsheet data can be turned into something genuinely useful for decision-making.

Learning Outcomes

- Understood the importance of building a dedicated Date table and enabling proper time-intelligence in Power BI.
- Gained hands-on experience writing DAX measures (SUM, DISTINCTCOUNT, DIVIDE, CALCULATE, SAMEPERIODLASTYEAR) to create meaningful business metrics.
- Learned to design and format multiple interactive visuals — KPI cards, line charts, bar charts, scatter charts, maps, and matrices.
- Practiced applying consistent themes, colors, and formatting to make a report look professional and presentation ready.
- Understood how to translate raw data patterns (like discount vs. profit) into clear, actionable business insights for decision-makers.
- Improved overall proficiency in using Power BI as an end-to-end tool — from data import to a finished, interactive business dashboard.